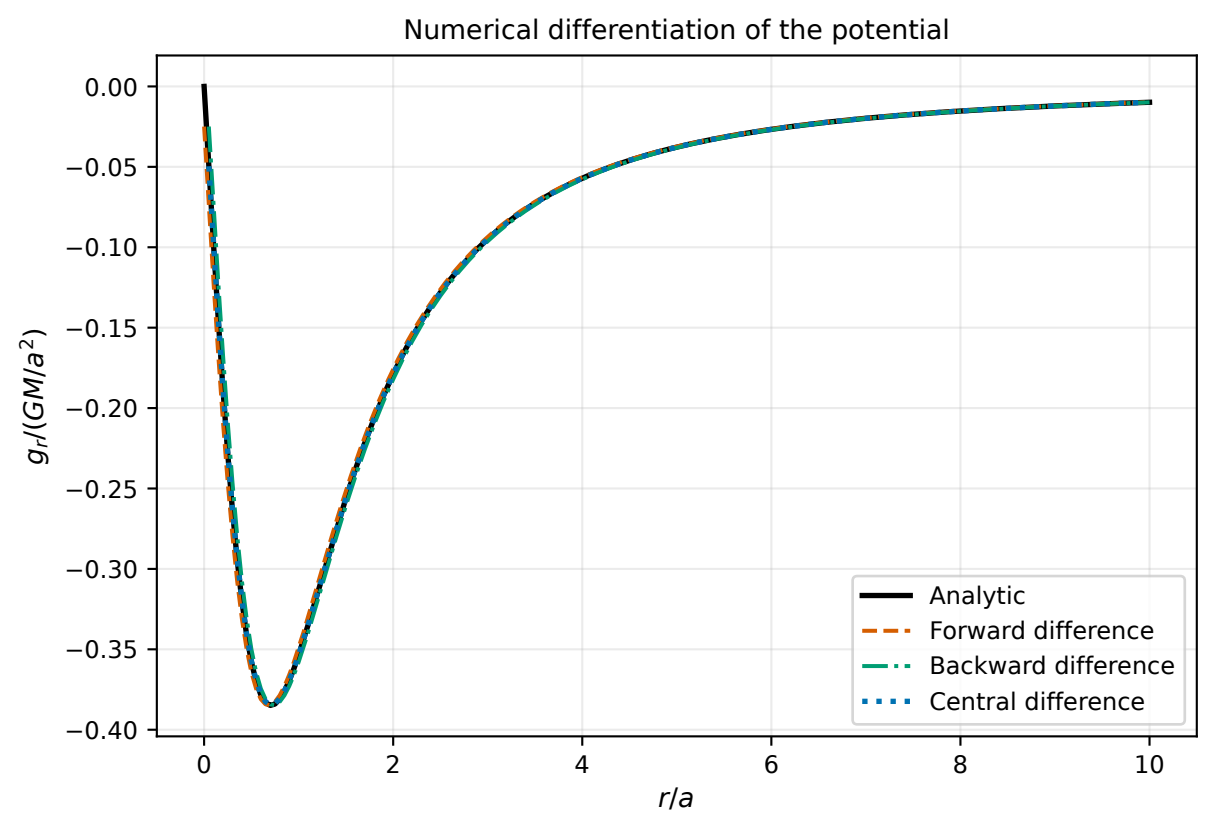
Numerical differentiation of the potential
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
−0.40
$g_r/(GM/a^2)$
0
2
4
6
8
10
r/a
Analytic
Forward difference
Backward difference
Central difference